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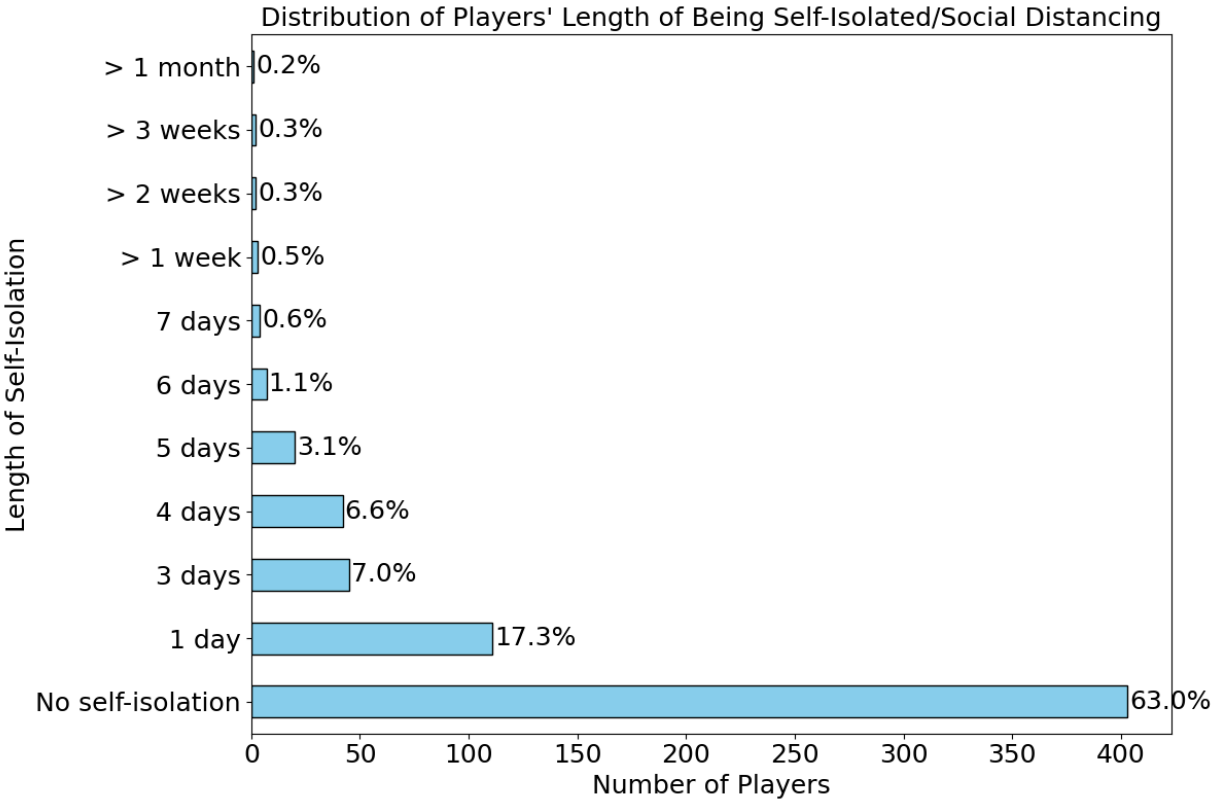
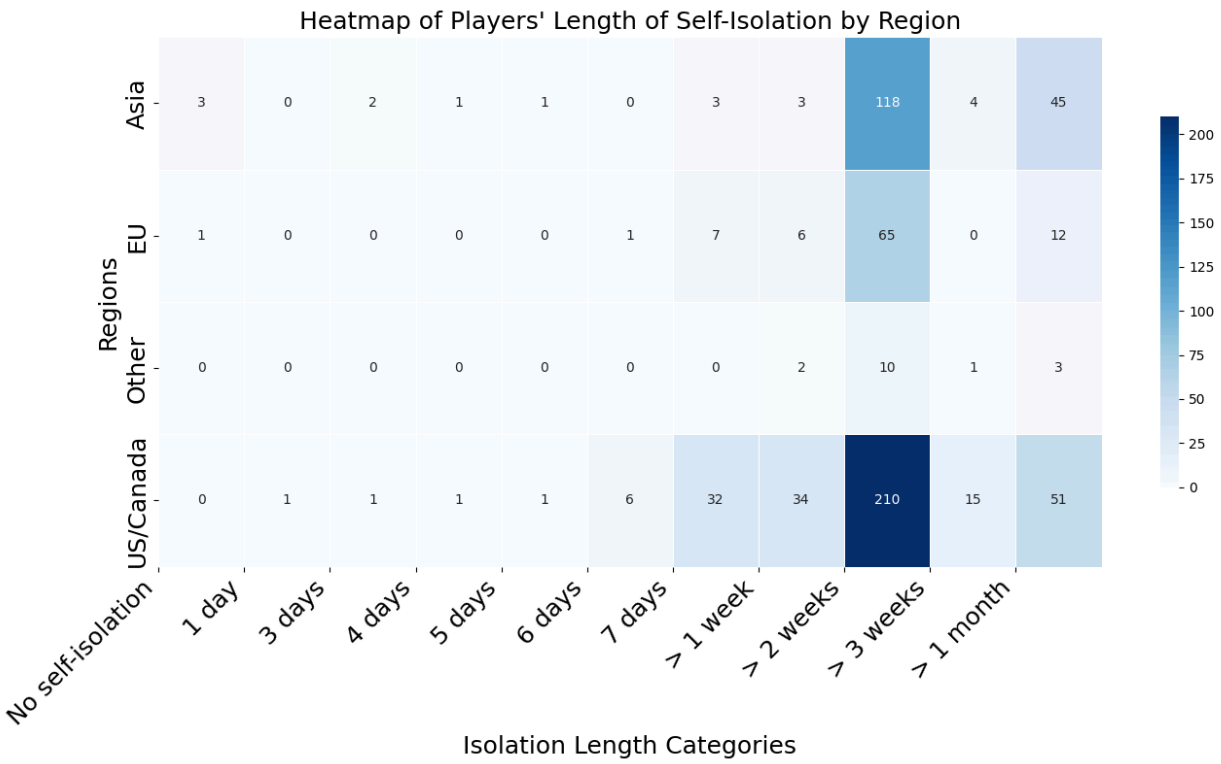
INTRODUCTION
This coursework delves into how social isolation influences in-game behaviors in Animal Crossing: New Horizons (ACNH), using a global dataset collected during the COVID-19 pandemic. The project is structured into three parts: Part A focuses on exploring data to uncover patterns in isolation lengths, regional differences, and gaming habits; Part B highlights the key behaviors that reflect the duration of isolation through feature selection techniques; and Part C applies machine learning to predict isolation lengths based on behavioral data, with supporting visualizations to present the findings. By combining data analysis, behavioral insights, and predictive modeling, the coursework provides a deeper understanding of how gaming served as a coping mechanism during times of social isolation.

DETAILED ANALYSIS
PART A
a. The horizontal bar chart effectively represents the distribution of players' self-reported durations of self-isolation or social distancing. It is highly effective as it displays both percentages and participant counts while organizing the periods from shortest to longest, allowing key patterns to be easily identified. The data for this analysis was extracted from B2.
A majority of players (63%) reported isolating for "More than a month", indicating prolonged adherence to these measures. In contrast, 17.3% reported no self-isolation or social distancing. Durations between "More than a week" and "More than three weeks" account for 3.1% to 7%, while very short durations (1-7 days) are minimal, each representing less than 1.5%. The chart highlights a strong tendency toward longer isolation durations, reflecting personal caution or compliance with public health mandates during the data collection period.

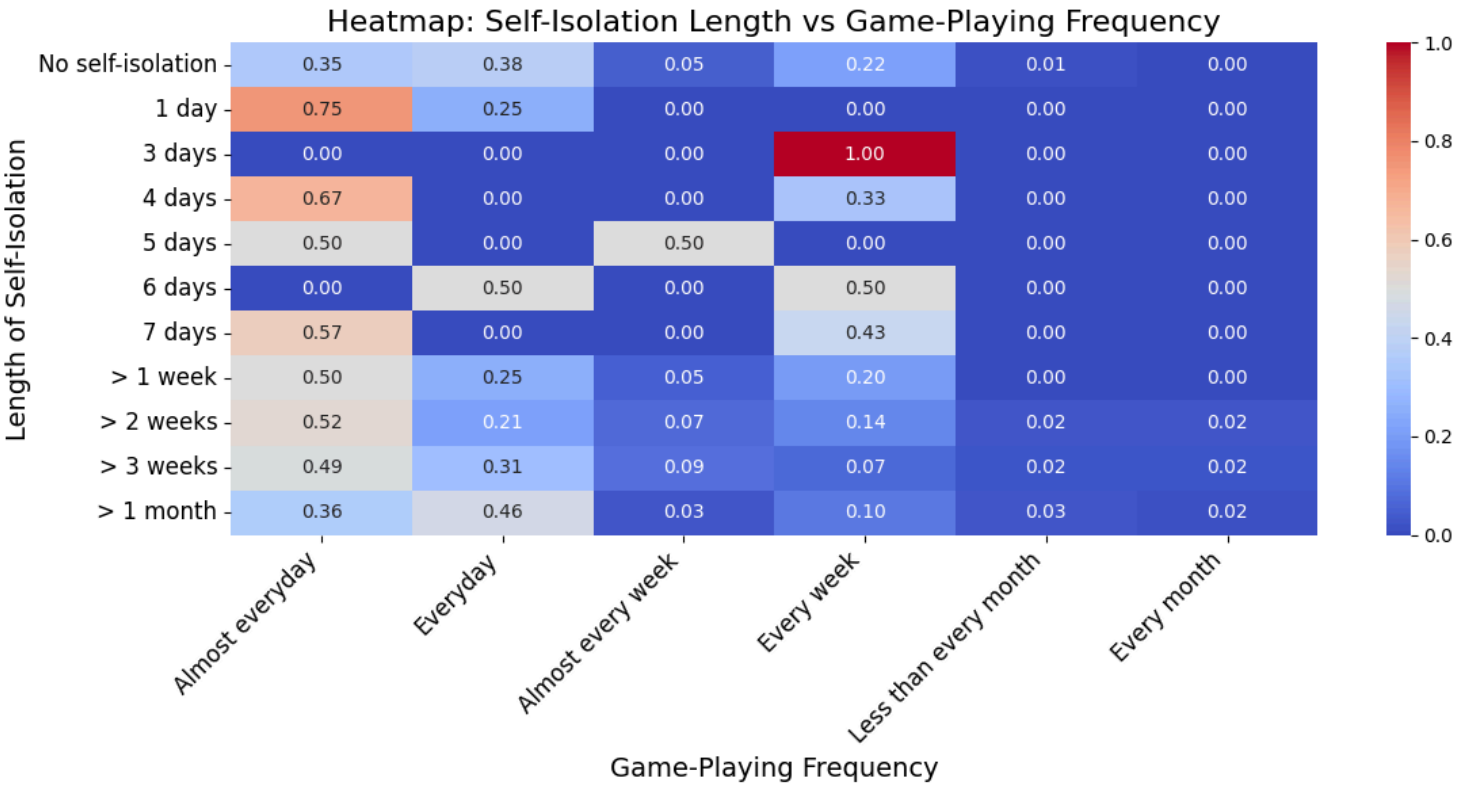


b. The heatmap provides a distinct pattern of the length of players' self-isolation by region. In the US/Canada, the self-isolation periods increase gradually across all the categories, with relatively low numbers for shorter durations such as "1 day" and "3 days", steadily rising through "7 days", "More than a week", and peaking at "More than a month". This suggests a consistent and progressive adherence to longer isolation periods in this region.
In contrast, Asia shows a different trend. The number of players in self-isolation remains relatively low across shorter isolation periods and shows a sudden increase in the "More than a month" category. This contrast between gradual and sudden increases provides insight into regional differences in isolation guidelines and government regulations during the pandemic. The EU shows minimal counts across all categories, possibly due to limited data collection from this region or a lower proportion of participants following structured isolation periods.

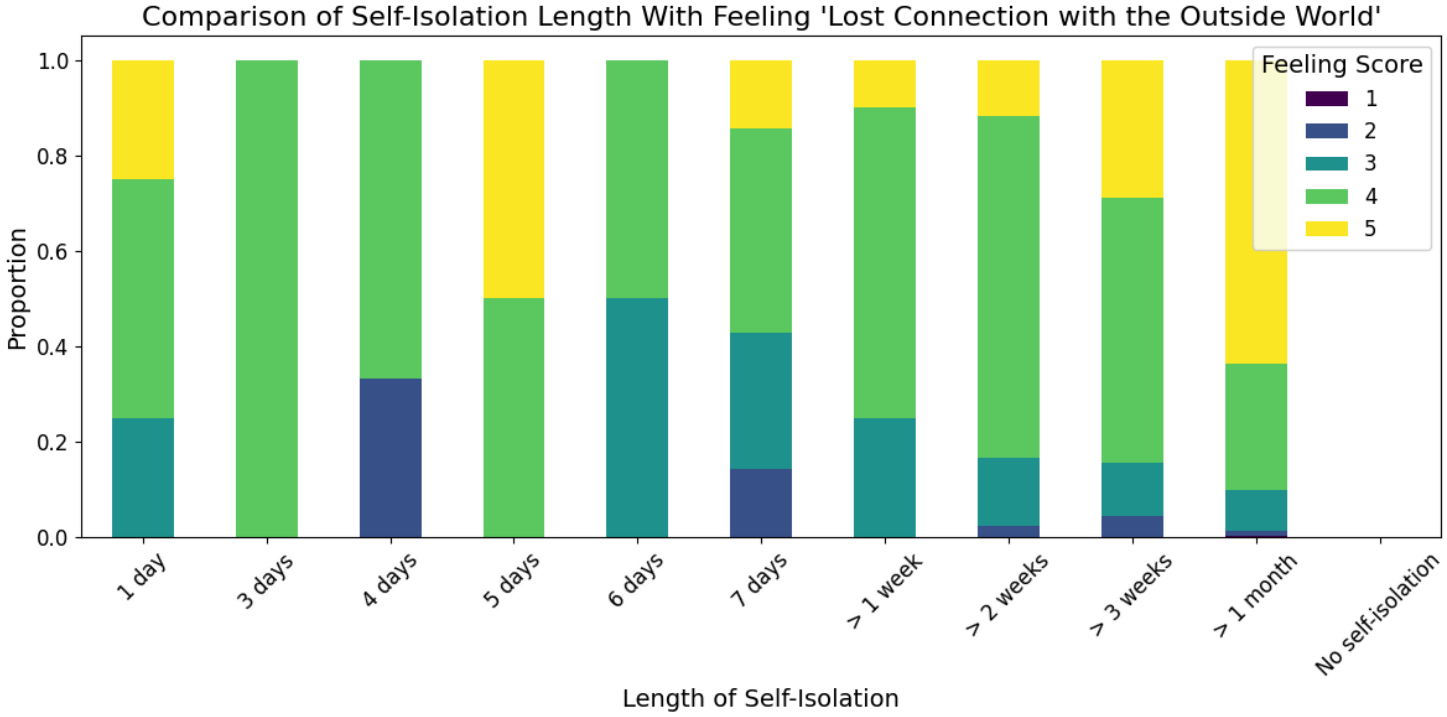


c. The heatmap effectively illustrates the relationship between players' length of self-isolation and their game-playing frequency. It pinpoints the exact relationship between data sourced from B2 (self-isolation duration) and D3 (game-playing frequency). The heatmap shows that frequent gamers, such as those who play "Every day" or "Almost every day", are more likely to maintain or increase their gaming during self-isolation. For instance, the highest score is seen among players isolating for 3 days and playing every day, achieving a perfect score of 1. Additionally, players who play "Almost every day" show consistent engagement across all isolation durations, reflecting a sustained habit.

In contrast, individuals who rarely play games ("Every month" or "Less than every month") show minimal engagement, even during extended isolation periods like "More than a month", indicating that self-isolation does not significantly encourage gaming behaviour among non-habitual players. Interestingly, participants who reported "No self-isolation/social distancing" exhibit moderate gaming habits, with a notable proportion playing "Every day". Overall, the heatmap suggests that while self-isolation does not necessarily initiate gaming behaviour, it may amplify gaming frequency for those already inclined to play regularly.



d. The stacked bar chart effectively visualizes the relationship between self-isolation duration and the feeling of "lost connection with the outside world", using a rating system from 1 to 5, where 5 indicates the highest disconnection. The chart's color scheme is accessible to individuals with color blindness, ensuring inclusivity. Data for this analysis was drawn from B2 (self-isolation duration) and F1 (game-related feelings). Players with longer isolation periods, such as "More than a month", show a significantly higher proportion of strong responses (score 5), which is more than double other scores for these durations, highlighting the heightened psychological impact of extended isolation. "5 days" stands out with a balanced 50/50 split between scores 4 and 5.
Shorter durations, like "1 day", show mixed responses with scores of 3, 4, and 5, while "3 days" reveals unanimous responses with a score of 4. Categories like "More than a week", "More than 2 weeks" and "More than 3 weeks" lean towards higher scores (4 and 5), reflecting stronger feelings of disconnection. Individuals with "No self-isolation/social distancing" display a more balanced mix of responses, but with notable instances of high scores (5). Overall, the chart underscores how prolonged isolation correlates with greater feelings of disconnection, emphasizing the emotional toll of extended self-isolation.



PART B

The primary objective was to identify the in-game behaviours most indicative of the length of self-isolation or social distancing. To achieve this, a machine learning pipeline was developed and utilised, leveraging the capabilities of the CatBoostClassifier after multiple model comparisons done in part c of this coursework. During feature selection, Behavioural features (E1, E2, ..., E28) were extracted from the dataset, as these columns represented in-game actions and activities that could potentially correlate with self-isolation duration. The target variable (B2), which originally contained descriptive labels for the length of self-isolation (e.g., "1 day", "More than 2 weeks"), was converted into numeric values using a mapping scheme. The values were subsequently grouped into three categories: Short, Medium, and Long, providing a more interpretable classification task.

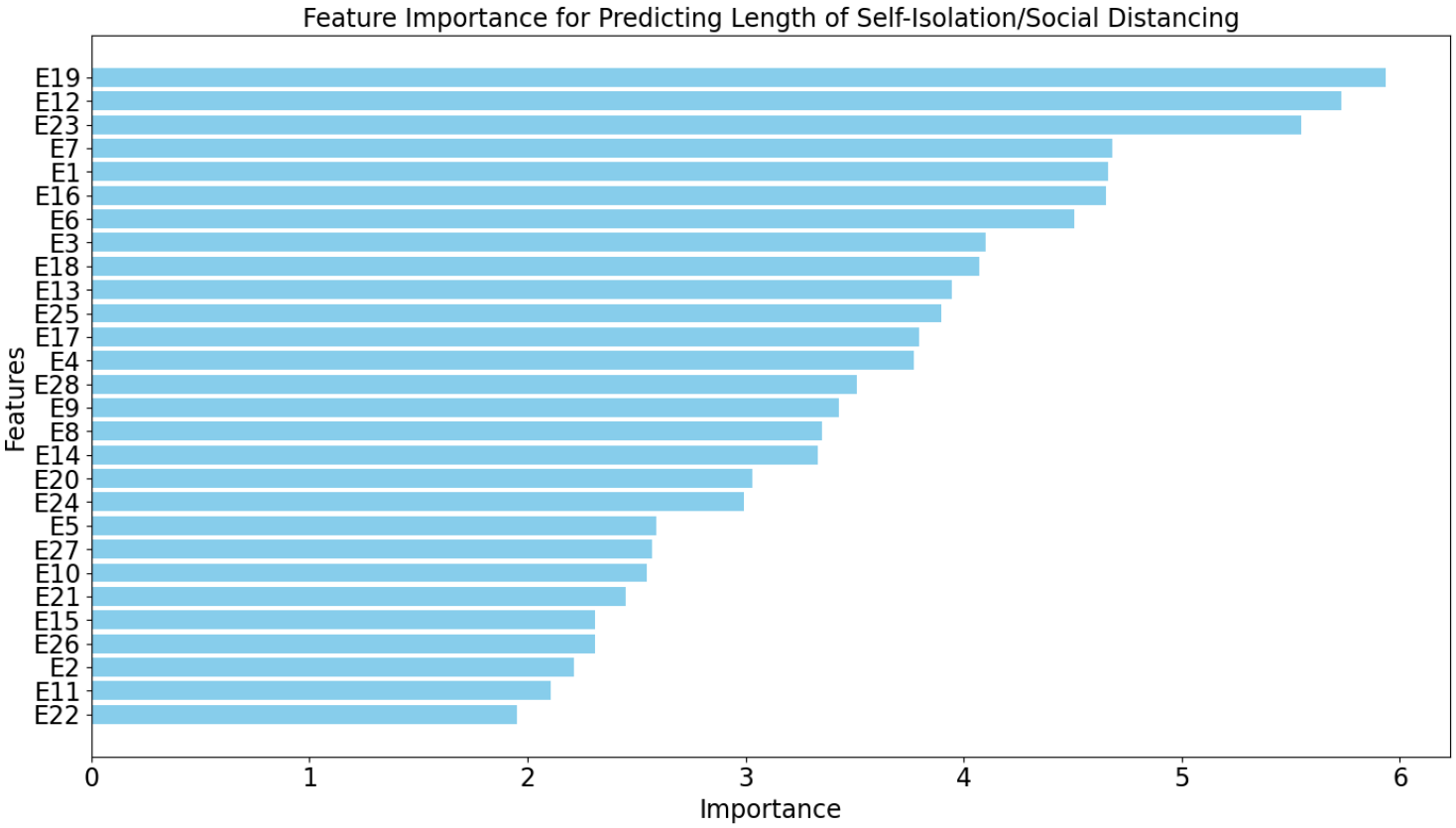
Addressing Class Imbalance (Using SMOTE)[1]:

The dataset exhibited a significant imbalance in the target classes, with shorter durations being overrepresented compared to longer durations. To mitigate this issue, the Synthetic Minority Oversampling Technique (SMOTE) [1] was applied, which generated synthetic samples for underrepresented categories, ensuring a balanced dataset. This approach was critical in allowing the model to learn meaningful patterns across all target classes without bias. While CatBoost was used as the classifier, as explained in detail in Part (c), SMOTE played a pivotal role in mitigating the imbalance, enabling the analysis to focus on the relationship between in-game behaviors and self-isolation durations.

Model Selection and Evaluation of Feature Importance:

After experimenting with various models such as Random Forest, Balanced Random Forest, XGBoost, Logistic Regression, LightGBM and Stacking Classifier; the CatBoostClassifier was chosen for its ability to handle

categorical data and imbalanced datasets effectively. Its built-in feature importance functionality ranked behavioral features by their contribution to predicting the target variable (B2_category), providing a quantitative measure of each feature's relevance. After training the CatBoostClassifier, feature importance scores were calculated. These scores indicate how much each in-game behavior contributed to the model's predictive accuracy. **E19 and E12** were identified as the most significant features, suggesting these activities were particularly popular among players with extended isolation periods. Features like **E23** and **E7** were also strongly correlated, possibly reflecting social or exploratory behaviors that resonate with players seeking connection or distraction during isolation.



PART C

The objective was to develop a machine learning model to predict self-isolation or social distancing durations based on in-game behavioral features. Framed as a multi-class classification problem, the target variable (B2_category) categorized durations (e.g., "1 day", "7 days", "More than 2 weeks") for finer granularity. The dataset was split into Training (60%), Validation (20%), and Test (20%) sets, with stratified sampling ensuring proportional representation. The preprocessing steps applied to the dataset ensured that the model received high-quality and balanced input data for training. One of the primary challenges was the imbalanced distribution of the target classes, with certain durations of self-isolation being significantly underrepresented. To address this, the SMOTE [1] algorithm was used as mentioned earlier in Part (b) of the coursework. SMOTE generated synthetic samples for the minority categories, such as longer durations of self-isolation, ensuring a more balanced dataset. This step was critical in preventing the model from becoming biased toward the majority classes and helped in capturing meaningful patterns across all categories.

Only the behavioural features (E1-E28) were included as predictors for the model based on the dataset. By narrowing down to these specific features, the preprocessing step ensured that the model concentrated on the in-game actions of players, which are hypothesized to correlate strongly with their self-isolation durations. This targeted approach helped improve both the interpretability and predictive power of the model. These preprocessing steps laid the foundation for the machine learning pipeline.

Model Selection and Tuning

The **CatBoostClassifier [6]** was chosen for its ability to handle challenges such as categorical features without manual encoding, simplifying workflows and reducing errors. While the dataset mainly contained numerical features (E1 to E28), this capability demonstrates CatBoost's flexibility for future mixed-data applications. Its minimal preprocessing requirements streamline the process and minimize encoding errors. Additionally, the **auto_class_weights [2]** parameter automatically adjusts class weights based on representation, making it effective for imbalanced target variables. This ensured the model accurately learned patterns for underrepresented

self-isolation durations without bias. Another key feature of CatBoost is its ability to compute feature importance scores, highlighting each feature's contribution to predictions. This was crucial for identifying in-game behaviours most predictive of self-isolation durations, enabling actionable insights into player behaviour.

Hyperparameter tuning for the CatBoostClassifier was conducted using **Optuna [3]**, a framework that efficiently searches the hyperparameter space with advanced techniques like **Tree-Structured Parzen Estimators (TPE) [4]**. Optuna automated the process, saving time while ensuring thorough exploration. During the tuning process, a 5-fold stratified cross-validation approach [5] was used to evaluate the model's performance for each hyperparameter configuration ensuring that the validation process was robust and unbiased, with all target classes proportionally represented in each fold. Stratification was particularly important due to the imbalanced dataset, as it prevented the validation results from being skewed by class imbalances. Several critical hyperparameters were optimized to maximize the model's predictive performance:

- **Iterations (500–2000):** Controls the number of trees, balancing underfitting and overfitting.
- **Tree Depth (6–10):** Determines the model's ability to capture complex patterns.
- **Learning Rate (0.01–0.3):** Controls weight updates; lower rates improve precision but require more iterations.
- **L2 Regularization (1–10):** Prevents overfitting, ensuring better generalization.
- **Bagging Temperature (0–1):** Introduces randomness in feature sampling, enhancing robustness.

Hyperparameter Optimization Results: Optuna identified optimal parameters for the **CatBoostClassifier**: Iterations: 1821, Depth: 9, Learning Rate: 0.145, L2 Regularization: 1.016, Bagging Temperature: 0.799. These settings balanced complexity (depth) with overfitting prevention (regularization and bagging temperature), ensuring high accuracy on unseen data.

Validation Set Performance: Achieved 100% precision, recall, and F1-score across all classes, indicating effective pattern learning from training data.

Original Dataset Performance: Accuracy of 97% showed consistency across categories, including imbalanced ones. A macro-average F1-score of 95% highlighted balanced class performance.

Test Set (Unseen Data) Performance: Accuracy: 94%.

Category	Precision (%)	Recall (%)	F1-Score (%)
0 (No self-isolation)	93	90	91
1 (1-2 weeks)	93	93	93
2 (More than 2 weeks)	96	99	97

Graphical Insights from Behavioral Trends

A line chart was chosen because it effectively visualizes changes and trends across durations, making it easier to compare feature behaviors and understand how player engagement evolves. The graph highlights distinct patterns for the top features. Players' in-game behaviors varied with isolation durations. Long isolations tend to focus more on strategic activities, such as reserving trees for bugs (E19), and immersive exploration such as collecting bugs (E7). Social activities, like sending gifts (E23), highlight the use of the game to maintain connections, especially during mid and long durations.

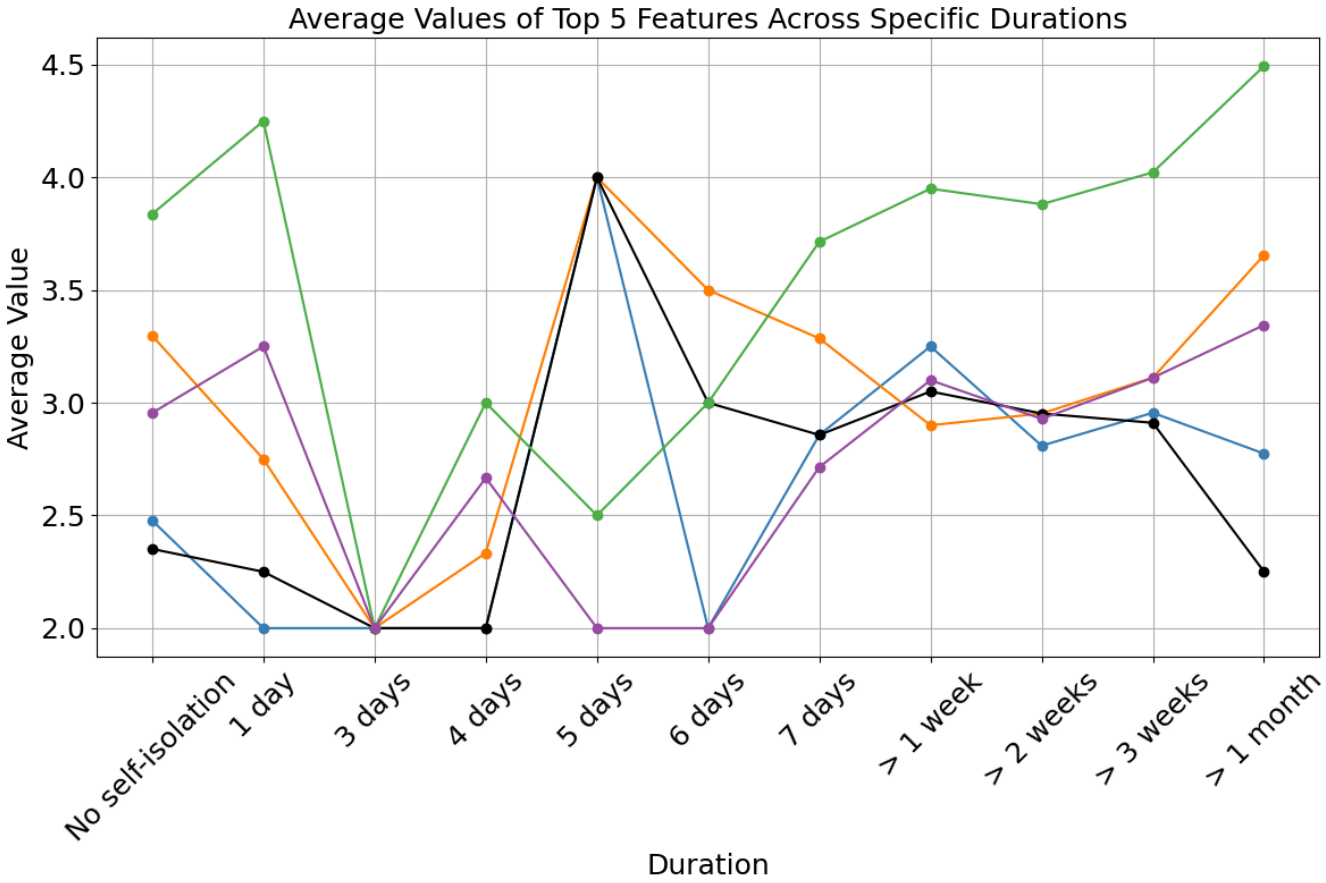
As isolation durations increase, players show a clear shift from casual, short-term tasks to more immersive and strategic gameplay. Shorter isolation periods are characterized by quick, goal-oriented actions such as catching bugs (E1) and selling bugs (E12), reflecting a focus on instant gratification. In contrast, longer isolations encourage players to engage in activities that require planning and patience, such as exploring or managing resources. Social engagement peaks during mid-length isolations, suggesting that players use the game to maintain connections during periods of moderate isolation.

This progression from casual to strategic and social gameplay reflects an adaptive approach, as players explore various in-game options to sustain interest and engagement. For example, while shorter isolations emphasize repetitive tasks for immediate rewards, longer isolations encourage a mix of immersive exploration and long-term goal planning, such as reserving trees (E19). Interestingly, certain features peak at specific durations, indicating players adapt their gameplay based on the stage of isolation. For instance, reserving trees (E19) sees a notable spike at around five days of isolation, while collecting bugs (E7) becomes increasingly prominent for durations

exceeding one month. These trends suggest players progressively transition from casual engagement to immersive activities as they settle into extended periods of isolation.

Catching bugs (E1) stands out as a steady activity across all durations, serving as a reliable and familiar task for players. This consistency highlights its importance as an "anchor" activity, providing a sense of routine and stability, particularly during periods of uncertainty. Additionally, players in longer isolations diversify their gameplay by balancing immersive exploration, strategic planning, and social activities, highlighting the game's ability to cater to various coping mechanisms and engagement levels. Social features like sending gifts (E23) play a crucial role during mid-length isolations, offering players a way to maintain connections and build a sense of community.

These trends demonstrate how isolation length affects players' focus on casual, social, or strategic gameplay, illustrating the game's versatility in supporting players through different phases of isolation. The ability to adapt gameplay to the length and nature of isolation highlights its value as both a source of entertainment and a coping mechanism.



CONCLUSION

This study demonstrates how social isolation influenced in-game behaviours in Animal Crossing: New Horizons during the COVID-19 pandemic. Through data analysis, feature selection, and machine learning, the findings give a clear connection between self-isolation durations and players' gaming habits. Players tend to present strategic and goal-focused activities when the isolation period is longer. Conversely, players focus more on casual and social activities in a shorter period. Using CatBoostClassifier, the analysis effectively predicted isolation durations based on in-game behaviours, highlighting gaming's role in maintaining connections and supporting mental well-being during longer social restrictions. These findings provide a basis for understanding gaming as a coping mechanism in times of crisis. The analysis also shows how the role of the game supports mental well-being during pandemics and offers a foundation for future research into gaming behaviours during crises.

Appendix

A. Dataset

Scidb.cn, A multinational data set of game players' behaviors and psychological factors, [online] [Accessed 1 Dec. 2024]. Available at: https://www.scidb.cn/en/detail?dataSetId=cb5d36cce29f4e5695a586c9b85d04b6

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B. Model

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C. Code Base - https://drive.google.com/file/d/1w6DaxuXBFCOWP3KiNM2gm9kpstCGUM-H/view?usp=sharing

D. Additional References

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Vuong, Q.H., Ho, M.T., La, V.P., Le, T.T., Nguyen, T.H.T. and Nguyen, M.H. 2021. A multinational data set of game players' behaviors in a virtual world and environmental perceptions. Data Intelligence. [Online]. 3(4), pp.606–630. [Accessed 1 December 2024]. Available from: https://doi.org/10.1162/dint_a_00111